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Microelectromechanical systems (MEMS): fabrication, design and applications

Jack W Judy

Electrical Engineering Department, University of California, Los Angeles 68-121
Engineering IV, 420 Westwood Plaza, Los Angeles, CA 90095-1594, USA

E-mail: jjudy@ucla.edu

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Abstract

Micromachining and micro-electromechanical system (MEMS) technologies can be used to produce complex structures, devices and systems on the scale of micrometers. Initially micromachining techniques were borrowed directly from the integrated circuit (IC) industry, but now many unique MEMS-specific micromachining processes are being developed. In MEMS, a wide variety of transduction mechanisms can be used to convert real-world signals from one form of energy to another, thereby enabling many different microsensors, microactuators and microsystems. Despite only partial standardization and a maturing MEMS CAD technology foundation, complex and sophisticated MEMS are being produced. The integration of ICs with MEMS can improve performance, but at the price of higher development costs, greater complexity and a longer development time. A growing appreciation for the potential impact of MEMS has prompted many efforts to commercialize a wide variety of novel MEMS products. In addition, MEMS are well suited for the needs of space exploration and thus will play an increasingly large role in future missions to the space station, Mars and beyond.

(Some figures in this article are in colour only in the electronic version)

1. Introduction

1.1. Revolutions at microscopic scales with macroscopic impact

The miniaturization of electrical circuits and systems continues to fuel a technological revolution responsible for a \$200B integrated-circuit (IC) industry, which has fundamentally changed the world economy and the way our society lives and works. For example, many products created by the IC industry (e.g., microprocessors, DRAM, FPGA, ASIC etc) enable the inexpensive production of extremely useful and popular electronic systems (e.g., personal computers, computer networks, instrumentation, cell phones, sophisticated electronic appliances etc).

The miniaturization of nearly all other types of device and system is arguably an even greater opportunity for com-

mercial profit and beneficial technological advances (e.g., micromechanical, microfluidic, microthermal, micromagnetic, microoptical and microchemical) [1]. However, instead of the traditional evolutionary engineering effort to reduce size and power while simultaneously increasing the performance of such a diverse set of systems, the field of microelectromechanical systems (MEMS) represents an effort to *radically* transform the scale, performance and cost of these systems by employing batch-fabrication techniques and the economies of scale successfully exploited by the IC industry [2]. Specifically, MEMS technology has enabled many types of sensor, actuator and system to be reduced in size by orders of magnitude, while often even improving sensor performance (e.g. inertial sensors, optical switch arrays, biochemical analysis systems etc) [3–6].

1.2. Acronyms

Due to the enormous breadth and diversity of the devices and systems that are being miniaturized, the acronym MEMS is not a particularly apt one (i.e., the field is more than simply micro, electrical and mechanical systems). However, the acronym MEMS is used almost universally to refer to the entire field (i.e., all devices produced by microfabrication except ICs). Other names for this general field of miniaturization include *microsystems technology* (MST), popular in Europe, and *micromachines*, popular in Asia.

1.3. Scaling advantages and issues

When miniaturizing any device or system, it is critical to have a good understanding of the scaling properties of the transduction mechanism, the overall design, the materials and the fabrication processes involved. The scaling properties of any one of these components could present a formidable barrier to adequate performance or economic feasibility. Due to powerful scaling functions and the sheer magnitude of the scaling involved (i.e., MEMS can be more than 1000 times smaller than their macroscopic counterpart), our experience and intuition of macroscale phenomena and designs will not transfer directly to the microscale.

1.3.1. Influence of scaling on material properties. When designing microfabricated devices, it is important to be aware that the properties of thin-film materials are often significantly different from their bulk or macroscale form. Much of this disparity arises from the difference in the processes used to produce thin-film materials and bulk materials. An additional source of variation is the fact that the assumption of homogeneity, commonly used with accuracy for bulk materials, becomes unreliable when used to model devices that have dimensions on the same scale as individual grains and other microscopic fluctuations in material properties. Thus, local changes in grain size and other characteristics could significantly alter the performance of MEMS produced either together (i.e., in one batch) or from batch to batch. One potential advantage of scaling MEMS to densities approaching the defect density of the material is that devices can be produced with a very low total defect count. This is one reason why the reliability of some MEMS, particularly those of simple mechanical design (e.g., cantilevers), can have better reliability than macroscopic versions [7]. However, due to the high surface-to-volume ratio of MEMS, more attention must be paid to controlling their surface characteristics.

Important material properties to characterize include elastic modulus, Poisson's ratio, fracture stress, yield stress, residual in-plane stress, vertical stress gradient, conductivity etc. Due to the flexibility of microfabrication, it is typically convenient to *integrate* microstructures that can be used to provide *in situ* measurements of material properties [8–10]. Many such microstructures have been used to reveal that thin-film material properties can vary tremendously from film to film without careful process control [11–13]. In fact, any high-precision and high-reliability MEMS application requires that significant effort be directed toward quantifying the precise material properties of the films being employed.

1.3.2. Scaling mechanical systems. From common experience we have all observed that small insects can survive a fall from a great height without significant damage and are capable of lifting objects many times their size or weight. This is due in part to the fact that mass is proportional to the volume of an object. When the linear dimensions of an object are reduced by a factor of s , the volume and hence the mass of the object is reduced by a factor of s^3 . However, when a mechanical flexure (e.g., cantilever beam) is scaled down by a factor s , its mechanical stiffness k ,

$$k = \frac{w \cdot t^3 \cdot E}{4L^3}, \quad (1)$$

with beam width w , thickness t , length L and elastic modulus E , is only scaled down by a factor of s [14]. Clearly the mechanical strength of an object is reduced much more slowly (s) than the inertial force it can generate (s^3). A beneficial consequence of this scaling characteristic is that MEMS can withstand tremendous accelerations without breaking or even being significantly disturbed. One extreme example is the fact that a micromechanical accelerometer survived being fired from a tank (i.e., experiencing more than a $\sim 100\,000\,g$ acceleration) even though the package and surrounding components, all of larger scale, did not fare as well. A negative consequence of the diminishing significance of inertial forces on the micrometer scale is that devices requiring proof masses (e.g. accelerometers) must have motion-detection systems with a much higher sensitivity.

1.3.3. Scaling fluidic systems. The dynamics of fluids in microscale systems is another example of how inadequate our macroscale experience is for predicting microscale behavior. The Reynolds number, which is a measure of flow turbulence (e.g., $Re < 2000$ representing laminar flow and $Re > 4000$ representing turbulent flow), is a function of the scale of the fluidic system, as shown in

$$Re = \frac{\rho \cdot V \cdot D}{\mu} \quad (2)$$

with density ρ , characteristic velocity V , characteristic length or diameter D and viscosity μ [15]. It is not surprising that although we commonly observe turbulent and chaotic fluid flow in most macroscopic systems, fluid flow in microscopic systems is almost entirely dominated by laminar flow conditions (i.e., as the dimensions of the fluidic system are scaled down by s , Re will also be scaled down by s and thus fluid flow becomes more laminar on a microscale). In fact, because of this behavior it is very challenging to accomplish thorough mixing in microfluidic systems. Although this behavior is expected from equation (2), actually quantifying the overall behavior of fluids on the microscale is not adequately predicted by the existing constitutive equations [16]. Presently there are a number of efforts in the MEMS research and development community to improve our ability to model microfluidic systems [17, 18].

1.3.4. Scaling chemical and biological systems. The scaling of chemical systems is limited by a fundamental tradeoff between sample size and detection limit. Although it is

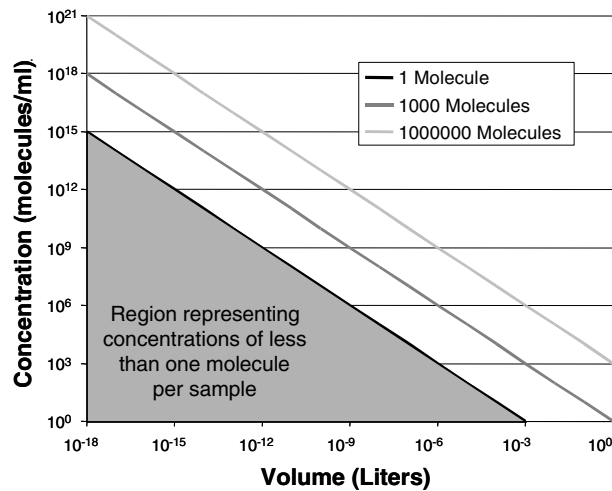


Figure 1. Tradeoff between sample size and detection limit.

typically advantageous to reduce the sample size, in a fixed concentration the total number of molecules that are available to be detected will also be reduced. Therefore, an increasingly sensitive detector will be needed but an obvious cut-off at detecting a single molecule is limiting. This tradeoff is illustrated in figure 1.

Most systems interfacing with biology are multidisciplinary (e.g. fluidic, electronic, mechanical etc) and thus the scaling properties of any of these components can limit the overall scaling of the system. The miniaturization of systems that interface with biology is also often limited by the application and the size of the relevant biological elements. For example, devices for manipulating cells can only be scaled down to cell-scale dimensions (e.g., typically 5–20 μm) whereas devices based on molecular function (e.g. DNA analysis) can be made considerably smaller. In addition, it has long been understood that microscopic biological organisms can overcome the detection-limit barrier, illustrated in figure 1, by using a gain mechanism (e.g., the generation of second-messenger molecules in response to the presence of a single target molecule [19]).

1.3.5. Scaling thermal systems. Some of the scaling properties of thermal systems can be easily predicted by analyzing the basic relationships involved. For example, as the linear dimensions of an object are reduced by s , the thermal mass of an object (i.e., the thermal capacity times the volume) will scale down more rapidly (s^3) than the rate of heat transfer (s^2). The result is that rapidly removing the heat from a microscale object is typically a simple matter since the heat can conduct in all directions (e.g. submersed in a fluid). However, since it is easy to microfabricate delicate structures that only allow heat conduction along paths of very high thermal resistance, it is also a simple matter to achieve very good thermal isolation (e.g., a device on a very thin membrane supported by long and narrow tethers made of a material with a high thermal resistivity).

A more careful analysis is needed to predict the thermal behavior of miniature structures when they are scaled down to sub-micron dimensions, the reason being that at these dimensions the structure and its elements are of the same

scale as the quantum mechanical phonon, or lattice vibrations, responsible for carrying heat energy. It is possible to construct sub-micron-scale devices where heat conduction can be significantly curtailed in a controlled fashion.

1.3.6. Scaling electrical and magnetic systems. Clearly the IC industry has shown that electrical systems, particularly circuits of resistors, capacitors, diodes and transistors, can be scaled tremendously with largely predictable behavior. However, a more careful analysis is needed for the case of electrostatic actuators. A figure of merit for actuators is the density of field energy U that can be stored in the gap between a rotor and stator.

For the case of electrostatic actuator the field energy density is

$$U_{\text{electrostatic}} = \frac{1}{2} \epsilon \cdot E^2 \quad (3)$$

with permittivity ϵ and electric field E . The maximum energy density of electrostatic actuators is limited by the maximum field that can be applied before electrostatic breakdown occurs. Macroscopically this maximum field is a constant ($\sim 3 \text{ MV m}^{-1}$) and the resulting energy density is only 40 J m^{-3} .

For magnetostatic actuators the field energy density is

$$U_{\text{magnetostatic}} = \frac{1}{2} \left(\frac{B^2}{\mu} \right) \quad (4)$$

with permeability μ and magnetic flux density B . The maximum energy density of magnetic actuators is essentially limited by saturation flux density B_{sat} , which is typically on the order of 1 T or 1 V s m^{-2} and the resulting energy density is $400\,000 \text{ J m}^{-3}$ (i.e., 10 000 times larger than $U_{\text{electrostatic}}$ for macroscopic devices).

Clearly, from the two cases above, we see that magnetic actuators can store many times more recoverable energy in the gaps between rotors and stators. Thus magnetic actuators dominate in the macroscopic world. This relative situation remains the same as devices are scaled down in size. However, as the air gap becomes smaller fewer ionization collisions happen and a larger field can be applied before a cascade electrostatic breakdown occurs. This trend continues until the gap is made small enough so that eventually a larger *voltage* must be applied in order for breakdown to occur. A plot of the breakdown voltage as a function of electrode gap, known as the Paschen curve, is given in figure 2 [20, 21].

The consequence for MEMS is that with gaps on the order $1 \mu\text{m}$, much larger voltages can be applied, that result in much larger electric fields and consequentially much larger energy densities. The gap at which the maximum *possible* energy density of electrostatic actuators exceeds that of magnetic actuators is shown in figure 3 to be $\sim 2 \mu\text{m}$. However, if reasonable voltages are considered, a much smaller gap will be needed to achieve the equivalent energy density of magnetic actuators (e.g., $\sim 0.05 \mu\text{m}$ for 10 V).

From figures 2 and 3 we see that the maximum energy density of magnetic actuators is not a function of air gap size. However, practical issues, such as resistive power losses and the integration of the necessary windings, are challenges to the extreme miniaturization of magnetic actuators. In addition, the size of magnetic domains (i.e., regions of material with uniform magnetization) is typically on the scale of micrometers

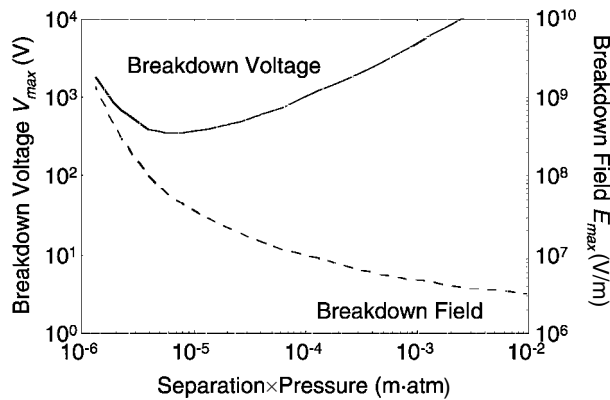


Figure 2. The Paschen curve [20, 21].

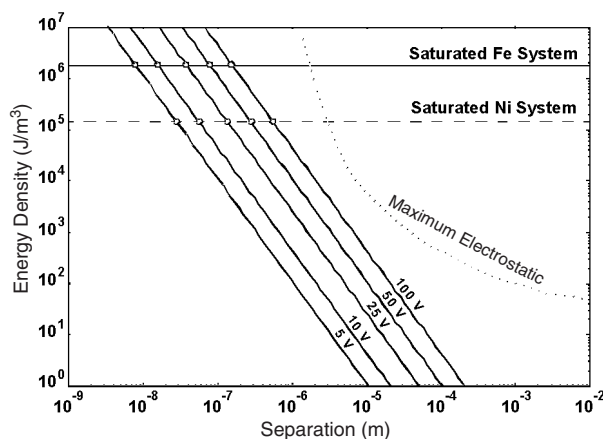


Figure 3. Comparison of electrostatic and magnetic energy densities as a function of rotor-stator gap.

in soft magnetic materials (e.g., NiFe), which are commonly used to produce magnetic MEMS. Therefore, the macroscopic assumption (i.e., the material consists of enough domains to ignore them individually and to only consider the ensemble average), will not be valid and new more complex models are needed for accurate and reliable prediction of experimental results. If magnetic MEMS are reduced to dimensions smaller than a typical domain, then the behavior will be dominated by single-domain phenomena.

1.3.7. Scaling optical systems. Microfabrication techniques have already been used to produce miniaturized optical systems (e.g., LEDs, lasers, integrated waveguides, mirrors and diffraction gratings). Due to the size of the wavelength of visible light (e.g., typically near 650 nm for red to approximately 475 nm for blue), the dimensions of integrated optical components are typically not smaller than this value. The behavior of scaled optical components is well predicted by existing constitutive equations (i.e., Fresnel) [22].

1.4. MEMS applications

MEMS applications and markets begin where traditional IC applications and markets end. Specifically, microfabrication and MEMS technologies can provide a means to interface the digital electronic world, dominated by the IC, with the

analog physical world. Due to the wide variety of nonelectrical signals of interest in the physical world (for an exhaustive list see [23]), many different transduction mechanisms are needed to *transduce* physical signals into electrical signals (i.e., sensors), which can be processed by IC-enabled electronic systems, as well as from electric signals into physical signals (i.e., actuators). In addition, sometimes it is advantageous to link transduction mechanisms in series (e.g., convert a thermal signal first into a mechanical signal, then into an optical signal, and finally into an electrical signal). Furthermore these sensing and actuating mechanisms can be combined with electronics to form complete microsystems.

Commercially successful devices and systems that use microfabrication and MEMS technologies include many microsensors (e.g., inertial sensors, pressure sensors, magnetometers, chemical sensors etc), microactuators (e.g., micromirrors, microrelays, microvalves, micropumps etc), and microsystems (e.g., chemical analysis, sensor-feedback-controlled actuators etc). For a wide-ranging discussion of nearly all types of micromachined transducer, the interested reader is directed to the book by Kovacs [4].

Thus far, the most successful MEMS products exploit one or more of the following characteristics.

Advantageous scaling properties. Some physical phenomena perform much better or are more efficient when miniaturized to the micrometer scale.

Batch fabrication. With lithographic processes and batch fabrication the cost of producing one MEMS device is not much more than the cost to produce many MEMS devices.

Circuit integration. A tremendous value can be derived by integrating circuits with MEMS (e.g., pre-amplification of sensor signals, reduced electrical parasitics, local closed-loop control, smaller overall package and system etc); however, cost and complexity can be prohibitive.

1.5. MEMS market

The MEMS market, as with an appropriate acronym for this field, is difficult to clearly define due to its diversity. A plot of several market projections for the MEMS industry is given in figure 4. Notice the wide range in predicted values (e.g., year 2000 sales range from \$4B to \$30B). Some of this variation is due to the different definitions of MEMS and microsystems used by the surveyors. For example, the surveyor must decide whether inkjet printer heads and magnetic recording heads should be included. Although both are transducers and both are produced with nonstandard IC-fabrication processes, neither contains any moving parts. This decision will have a tremendous impact on the value of the microsystem market, since each is very large on its own (i.e., magnetic recording heads, \$4.5B in 1996 and \$12B in 2002, and inkjet printer heads, \$4.4B in 1996 and \$10B in 2002).

In addition to magnetic recording heads and inkjet printer heads, the existing market for microsystems is dominated by pressure sensors, inertial sensors, chemical sensors, *in vitro* diagnostics, infrared imagers and magnetometers. The future looks bright as new types of microsystem emerge in

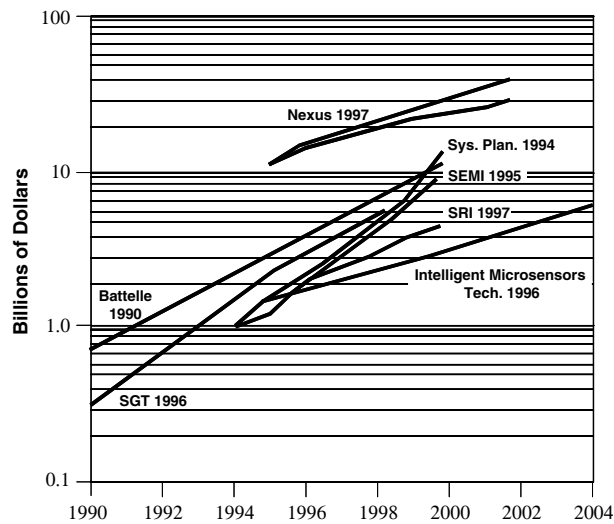


Figure 4. Market projections for microsystems [24].

products for additional markets (e.g., drug delivery systems, optical switches, chemical lab-on-a-chip systems, valves, RF switches, microrelays, electronic noses etc).

1.6. History of micromachining and MEMS

Any history of this field is dependent on the definition of a few key terms. For the purposes of this review, the following will be used.

Micromachining. Any process that deposits, etches or defines materials with minimum features measured in micrometers or less.

MEMS. All devices and systems produced by micromachining other than integrated circuits or other conventional semiconductor devices. Typically they have dimensions ranging from nanometers to centimeters.

The history of MEMS, as with its definition, is dependent on the development of micromachining processes.

1500. Early lithographic processes for defining and etching sub-mm features [3].

However, the micromachining processes with the greatest recent impact have been derived by those used to produce ICs. Key milestones in the development of IC micromachining are the following.

1940s. The development of pure semiconductors (Ge and Si), which was driven by the development of radar during World War II.

1947. The invention of the point-contact transistor, that heralded the beginning of the semiconductor circuit industry.

1949. The ability to grow pure single-crystal silicon improved the performance of semiconductor transistors, but their cost and reliability was still not completely satisfactory.

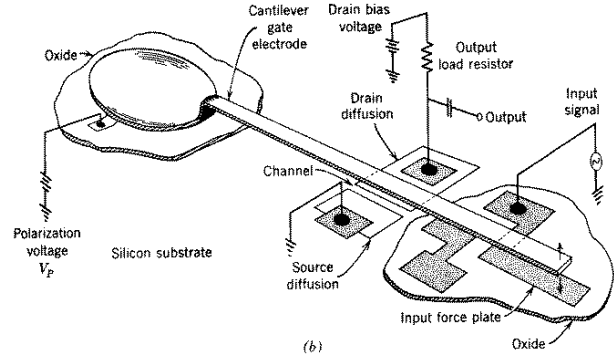


Figure 5. Resonant gate transistor [26].

1959. Professor Feynman gave his famous lecture titled 'There is plenty of room at the bottom' [25]. In it he described the enormous amount of space available on the microscale: 'The entire encyclopedia could be written on the head of a pin'. Thus he was describing the enormous potential of microfabrication on the eve of its invention. In addition, he was not satisfied with just miniaturizing information. He foresaw the miniaturization of machines and in fact famously challenged the world to 'fabricate a motor with a volume less than 1/64 of an inch on a side'.

1960. Invention of the planar batch-fabrication process tremendously improved the reliability and cost of semiconductor devices. In addition, the planar process allowed for the *integration* of multiple semiconductor devices onto a single piece of silicon (i.e., *monolithic integration*). This invention heralded the beginning of the IC industry. Although the early planar process produced relatively large devices ($> \text{mm}$), it was a tremendously scalable process that could micromachine an increasing number of devices.

1960. With the invention of the metal-oxide-semiconductor field-effect transistor (MOSFET), the IC industry embarked on a continuous effort to miniaturize increasingly complex circuits.

1964? The resonant gate transistor, produced by Nathenson at Westinghouse and shown in figure 5, was the first engineered batch-fabricated MEMS device [26]. The electrostatically driven motion of the cantilevered gold gate electrode modulates the electrical characteristics of the device.

1970. The development of the microprocessor, which found many applications that have been responsible for transforming our society, drove the demand for ICs even higher. The observation by Moore, that the number of transistors integrated onto a chip doubles every 18 months, has held true for the past 30 years.

1970s and 1980s. MEMS commercialization was started by several companies (e.g., IC Transducers, Foxboro ICT, Transensory Devices, IC Sensors and Novasensor) that produced parts for the automotive industry.

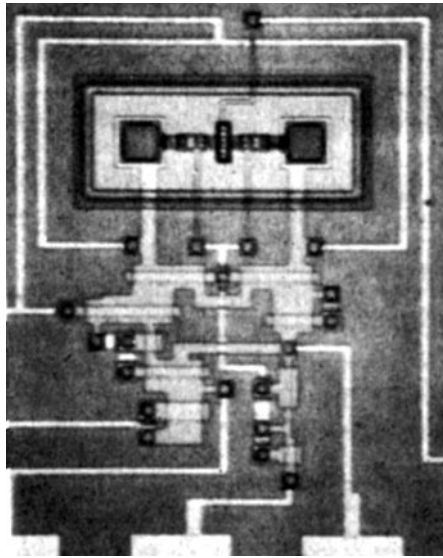


Figure 6. First polysilicon surface micromachined MEMS device integrated with circuits [29].

1982. Kurt Petersen's seminal paper titled 'Silicon as a mechanical material' discussed the development of many micromechanical devices and has been instrumental in increasing the awareness of the possibilities that MEMS has to offer [27].

1983. In a lecture titled 'Infinitesimal machinery', Professor Feynman reflected that his earlier miniaturization challenge was not difficult enough since it was accomplished by hand (i.e., not batch fabricated) by McLellan [28].

1984. Howe and Muller at the University of California, Berkeley (UCB) developed the polysilicon surface micromachining process and used it to produce MEMS with integrated circuits (figure 6) [29]. This technology has served as the basis for many MEMS products.

1989. Researchers at UCB and MIT independently developed the first electrostatically controlled micromotors that used rotating bearing surfaces [30–32]. Although no commercial product presently uses this micromotor technology, it served as a valuable technology driver for the field of MEMS.

1991. Microhinges developed at UCB by Pister *et al* [33] extended the surface micromachined polysilicon process so that large structures could be assembled out of the plane of the substrate, finally giving MEMS significant access to the third dimension.

1990s. A tremendous increase in the number of devices, technologies, and applications (too many to mention individually) has greatly expanded the sphere of influence of MEMS—and it continues today.

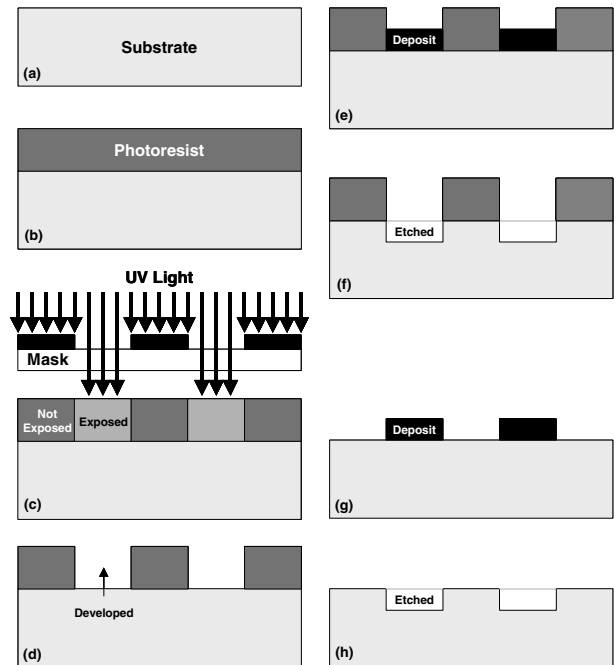


Figure 7. Photolithographic process.

2. Micromachining

Although many of the microfabrication techniques and materials used to produce MEMS have been borrowed from the IC industry, the field of MEMS has also driven the development and refinement of other microfabrication processes and materials not traditionally used by the IC industry.

Conventional IC processes and materials:

- photolithography, thermal oxidation, dopant diffusion, ion implantation, LPCVD, PECVD, evaporation, sputtering, wet etching, plasma etching, reactive-ion etching, ion milling [3];
- silicon, silicon dioxide, silicon nitride, aluminum.

Additional processes and materials used in MEMS:

- anisotropic wet etching of single-crystal silicon, deep reactive-ion etching (DRIE), x-ray lithography, electroplating, low-stress LPCVD films, thick-film resist (SU-8), spin casting, micromolding, batch microassembly [3];
- piezoelectric films (e.g., PZT), magnetic films (e.g., Ni, Fe, Co, and rare earth alloys), high-temperature materials (e.g., SiC and ceramics), mechanically robust aluminum alloys, stainless steel, platinum, gold, sheet glass, plastics (e.g., PVC and PDMS).

Of these processes and materials, photolithography is the single most important process that enables ICs and MEMS to be produced reliably with microscopic dimensions and in high volume. The essentials of the photolithographic process are illustrated in figure 7.

The process begins by selecting a substrate material and geometry. Typically a single-crystal silicon wafer, 4" to 8" in diameter, is used, although many other materials and geometries have also been used successfully (figure 7(a)).

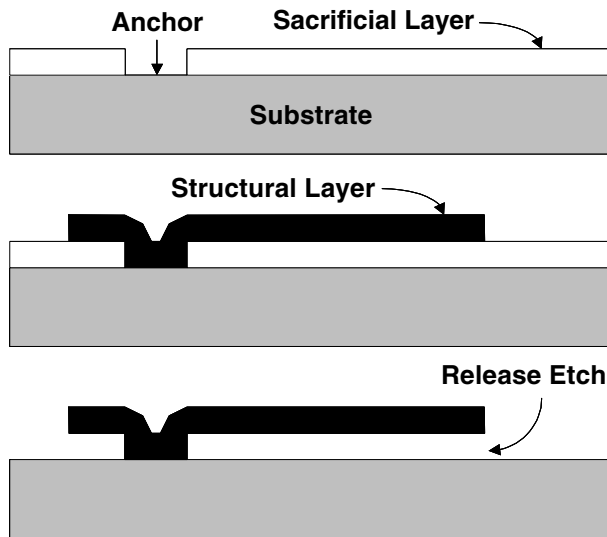


Figure 8. Surface micromachining and the sacrificial layer technique.

Next, the substrate is coated by a photosensitive polymer called a photoresist (figure 7(b)). A mask, consisting of a transparent supporting medium with precisely patterned opaque regions, is used to cast a highly detailed shadow onto the photoresist. The regions receiving an exposure of ultraviolet light are chemically altered (figure 7(c)). After exposure, the photoresist is immersed in a solution (i.e., developer) that chemically removes either the exposed regions (positive process) or the unexposed regions (negative process) figure 7(d). After the wafer is dried, the photoresist can be used as a mask for a subsequent deposition (i.e. additive process) figure 7(e), or etch (i.e., subtractive process) figure 7(f). Lastly, the photoresist is selectively removed, resulting in a micromachined substrate (figures 7(g), (h)). The permutations of materials and processes for depositing and etching makes it impossible to discuss them in sufficient detail. For a thorough treatment of deposition and etching processes, the interested reader is directed to the book by Madou [3].

The methods used to integrate multiple patterned materials together to fabricate a completed MEMS device are just as important as the individual processes and materials themselves. The two most general methods of MEMS integration are described in the next two sections: surface micromachining and bulk micromachining.

2.1. Surface micromachining

Simply stated, surface micromachining is a method of producing MEMS by depositing, patterning and etching a sequence of thin films, typically 1–100 μm thick. One of the most important processing steps required for dynamic MEMS devices is the selective removal of an underlying film, referred to as a *sacrificial layer*, without attacking an overlying film, referred to as the *structural layer*. Figure 8 illustrates a typical surface micromachining process [34]. Surface micromachining has been used to produce a wide variety of MEMS devices for many different applications. In fact, some of devices are produced commercially in large volumes (>2 million parts per month).

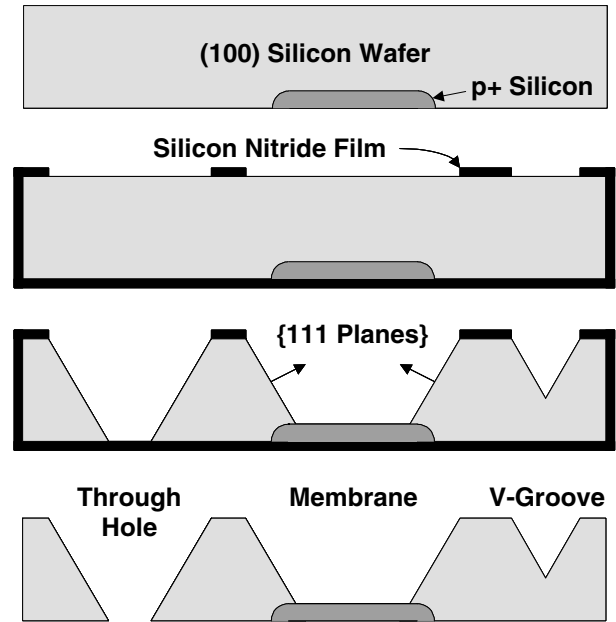


Figure 9. Bulk micromachining along crystallographic planes.

2.2. Bulk micromachining

Bulk micromachining differs from surface micromachining in that the substrate material, which is typically single-crystal silicon, is patterned and shaped to form an important functional component of the resulting device (i.e., the silicon substrate does not simply act as a rigid mechanical base as is typically the case for surface micromachining). Exploiting the predictable anisotropic etching characteristics of single-crystal silicon, many high-precision complex three-dimensional shapes, such as V-grooves, channels, pyramidal pits, membranes, vias and nozzles, can be formed [4,27]. An illustration of a typical bulk micromachining process is given in figure 9.

2.2.1. Deep reactive ion etching (DRIE). A dry etch process, patented by the Robert Bosch Corp [35], can be used to etch deeply into a silicon wafer while leaving vertical sidewalls and is independent of the crystallographic orientation (figure 10) [36]. This unique capability has greatly expanded the flexibility and usefulness of bulk micromachining.

2.2.2. Micromolding (HEXSIL). The combination of DRIE and conformal deposition processes, such as LPCVD polysilicon and silicon dioxide, can be used to create micromolded structures [37]. The process begins with a bulk-etched pattern in the silicon substrate by DRIE (figure 11(a)). Next, sequential conformal depositions are performed (e.g., SiO_2 , undoped polysilicon, doped polysilicon and plated nickel) (figures 11(b), (c)). Note that narrow trenches will be filled before wider trenches and thus the width can dictate the overall composition of materials in each trench. Access to the sacrificial SiO_2 is then achieved either by etching or polishing. Lastly, the sacrificial layer is removed and the microstructure that has been molded to the substrate is ejected and the process can repeat with the recycled substrate (figure 11(d)). With this process, thick microstructures (e.g., 500 μm thick) can

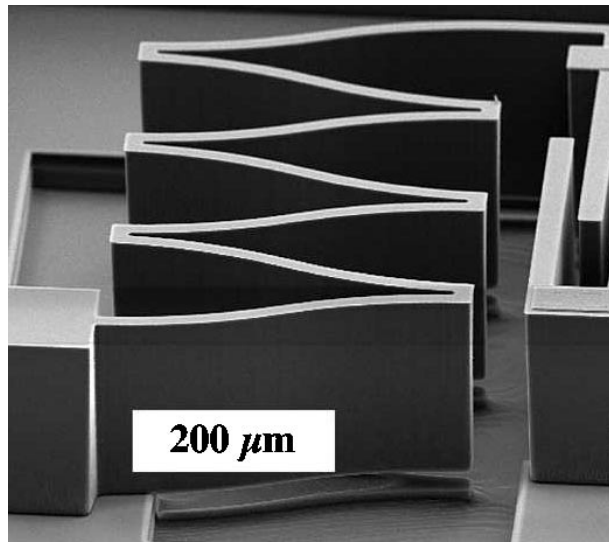


Figure 10. Microflexure created by vertical etching through a wafer with DRIE [36].

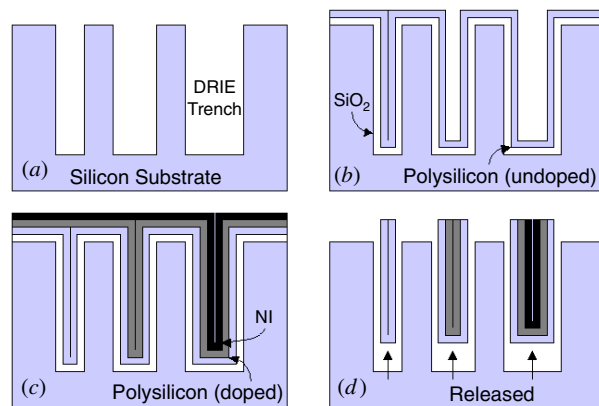


Figure 11. Micromolding process [37].

be realized with thin-film depositions and only one deep-etching step.

2.3. Substrate bonding

Silicon, glass, metal and polymeric substrates can be bonded together through several processes [3, 4] (i.e., fusion bonding [38], anodic bonding [39], eutectic bonding [40] and adhesive bonding [41]). Typically at least one of the bonded substrates has been previously micromachined either by wet etching with an anisotropic silicon etchant or dry etching by DRIE. Substrate bonding is typically done to achieve a structure that is difficult or impossible to form otherwise (e.g., large cavities that may be hermetically sealed, a complex system of enclosed channels) or simply to add mechanical support and protection.

2.4. Nonsilicon microfabrication

The development of MEMS has contributed significantly to the improvement of nonsilicon microfabrication techniques. Two prominent examples are LIGA and plastic molding from micromachined substrates.

2.4.1. LIGA. LIGA is a German acronym standing for *lithographie* (lithography), *galvanoformung* (plating) and *abformung* (molding) [42]. However, in practice LIGA essentially stands for a process that combines extremely thick-film resists (often >1 mm thick) and high-energy x-ray lithography (~ 1 GeV), which can pattern thick resists with high fidelity and also results in vertical sidewalls. Although some applications may require only the tall patterned resist structures themselves, other applications benefit from using the thick resist structures as plating molds (i.e., material can be quickly deposited into a highly detailed mold by electroplating). A drawback to LIGA is the need for high-energy x-ray sources (e.g., synchrotrons or linear accelerators) that are very expensive and rare—in the US only a few such sources exist.

2.4.2. SU-8. Recently a cheap alternative to LIGA, with nearly similar performance, has been developed. The solution is to use a special epoxy-resin-based optical resist, called SU-8, that can be spun on in thick layers (>500 μm), patterned with commonly available contract lithography tools and yet still achieve vertical sidewalls [3, 43].

2.4.3. Plastic molding with PDMS. Polydimethylsiloxane (PDMS) is a transparent elastomer that can be poured over a mold (e.g., a wafer with a pattern of tall SU-8 structures), polymerized and then removed simply by peeling it off of the mold substrate [44]. The advantages of this process include (1) many inexpensive PDMS parts can be fabricated from a single mold, (2) the PDMS will faithfully reproduce even sub-micron features in the mold, (3) PDMS is biocompatible and thus can be used in a variety of BioMEMS applications and (4) since PDMS is transparent, tissues, cells and other materials can be easily imaged through it. Common uses of PDMS in biomedical applications include microstamping of biological compounds (e.g., to observe geometric behavior of cells and tissues) and microfluidic systems [44–46].

2.5. Integration of circuits

As noted before, the integration of circuits can greatly improve the performance of many MEMS. However, this does not come without a price. The microfabrication processes for ICs are relatively long, complex and costly when compared with many MEMS fabrication processes. To combine MEMS with ICs requires much careful consideration of the manufacturing feasibility, complexity, reliability, yield and cost. The questions are the following. Should MEMS and ICs be monolithically integrated or separately produced and assembled together? If integrated together, should the MEMS be fabricated on the substrate before or after the ICs, or should the fabrication of both be interleaved together? The following is a brief summary of the advantages and disadvantages of each approach:

2.5.1. Fabricate separately and then assemble. This method has the lowest fabrication cost but the assembly method can significantly reduce the performance (i.e., higher parasitics) and reliability (i.e., more points of failure) and substantially increase cost.

2.5.2. Monolithic integration of ICs first. The low processing temperature ceiling ($\sim 400^\circ\text{C}$) imposed by the materials (e.g., melting point of aluminum) and electrical characteristics (i.e., movement of carefully designed dopant distributions) severely limits the maximum processing temperature for the subsequent steps used in the MEMS fabrication process (e.g., LPCVD polysilicon is deposited at $\sim 600^\circ\text{C}$). The yield of the MEMS process must be very high to conserve the expense associated with the IC process.

2.5.3. Monolithic integration of MEMS first. Although most MEMS often do not have a significant temperature ceiling and represent a lower investment of resources before the high-yield IC process begins, MEMS processes typically result in a substrate with a large surface topology (e.g., several micrometers). An IC process with small features requires a highly planarized substrate to obtain reasonable yields. This issue can be addressed by adding additional processing steps, at additional cost, to planarize the MEMS substrate before IC fabrication begins [47]. A significant advantage is that any IC fabrication process could be used since many MEMS materials and structural characteristics are far less temperature sensitive.

2.5.4. Monolithic integration of both MEMS and ICs in a mixed process. By interleaving the processing steps for the MEMS and IC components, a shorter process is possible but the development time is considerably longer. Also the IC fabrication facility used must be willing to allow nonstandard fabrication steps—something typically avoided in the IC industry to maximize yield. Despite the difficulties, Analog Devices, Inc. produces their commercially successful MEMS products (i.e., inertial sensors) using this method.

2.6. Foundries

Unfortunately, the cost of a microfabrication facility capable of producing MEMS is often prohibitively expensive for many companies, universities and organizations. In order to maximize the number of people working in the field of MEMS, some microfabrication facilities make their processes publicly available for a modest fee. These MEMS *foundries* have had a tremendously positive impact on the field as a whole, despite the fact that the foundry process and the materials are rigidly defined (i.e., the order, thickness and composition of any of the layers in the fabrication process cannot be changed). The most prominent MEMS foundries include the MUMPS process by Cronos [48], the SUMMIT process by Sandia National Laboratories [49], the iMEMS process by Analog Devices (figure 12) [50] and the IC foundry broker MOSIS [51].

3. Micromechanisms

Most MEMS consist of some combination of a few basic building blocks or micromechanisms. The following is a brief list of some of the more prominent micromechanisms:

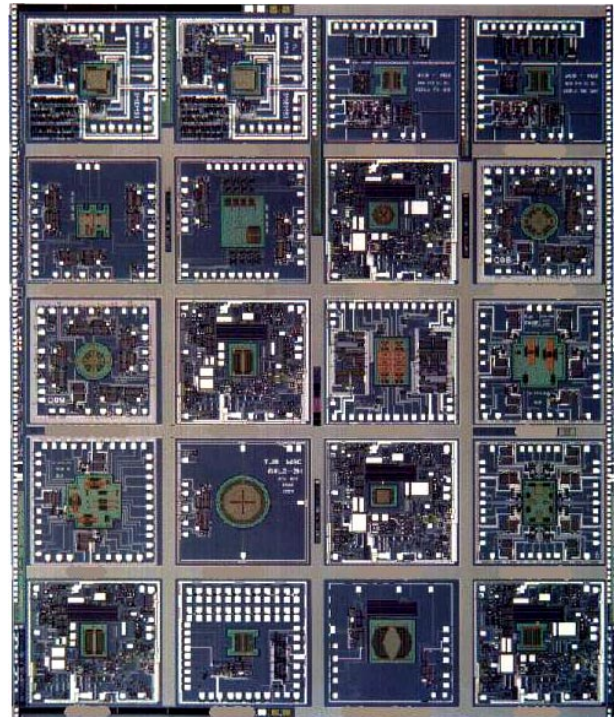


Figure 12. Photograph of several dice produced with the Analog Devices iMEMS foundry process (note that each chip represents a different design submitted by a different user).

3.1. Pits, grooves and channels

By combining photolithography and either deep-etching (e.g., bulk micromachining, KOH etching or DRIE) or thick-deposition processes (e.g., plating, PDMS micromolding and thick-film lithography) it is a simple matter to form deep pits, grooves and channels of many geometries. These micromechanisms are a building block for MEMS applications that need to contain or confine relatively large objects (e.g., optical fibers, relatively large quantities of fluid, high throughput flows and biological cells) [4,27].

3.2. Microflexures

The earliest microdynamic MEMS used microflexures to achieve consistent movement on a microscale [26]. Although many different and highly complex microflexure systems have been constructed, most MEMS simply use combinations of the most basic flexural elements: cantilever beams, bridges, torsion bars, plates and membrane. It should be noted that microflexures can have *extreme* aspect ratios (e.g., a freestanding cantilever with length $1000\ \mu\text{m}$, width $10\ \mu\text{m}$ and thickness $1\ \mu\text{m}$) that are rare in macroscopic systems.

The production of extremely compliant flexures can be accomplished if extra care is taken during their production. In particular, the release etch must be very gentle so that the flexure does not become broken or stuck to the substrate or other objects nearby. Since wet etchants can generate high fluid flow and surface tension forces, efforts are made to make the surfaces of the flexure and the substrate hydrophobic [52] or a *dry* release etch (e.g., plasma etching, XeF_2 etc) is used since it does not have a fluid meniscus and will not generate surface tension forces [53].

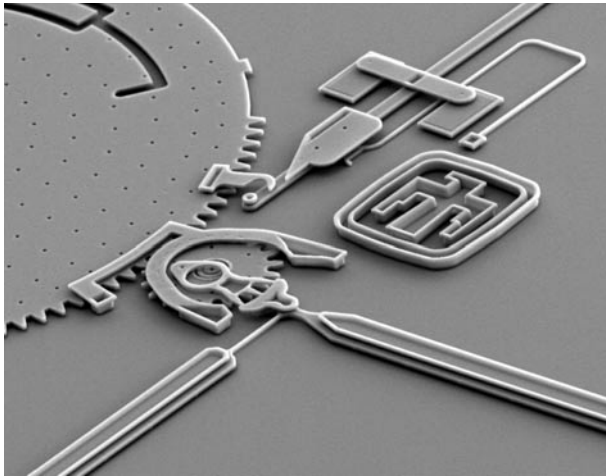


Figure 13. MEMS with rotary bearing surfaces and interlocking gears (Sandia National Laboratories) [55].

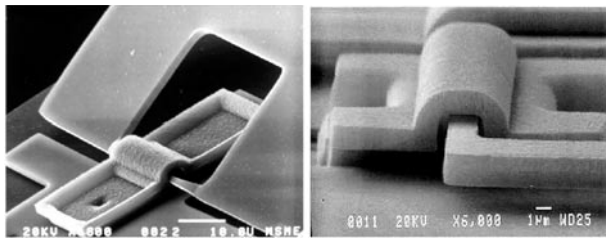


Figure 14. Hinged microstructures [33].

3.3. Microbearing surfaces

To enable fully free structures capable of unlimited rotation or translation, microbearing surfaces are needed (e.g., bearing hubs and sliders respectively). In-plane rotary hubs enable the development of micromotors and complex gear trains (figure 13) [54, 55]. Out-of-plane hinges enable the development of tall microstructures that make effective use of the space above the chip (figure 14) [33, 56]. A serious issue with microbearing surfaces is that the amount of mechanical slop is a large percentage of the size of the bearing elements (i.e., the relative tolerances in MEMS are typically much worse than that easily achieved with conventional machining techniques—approximately 10 and 0.1% respectively) [3]. Due to poor relative tolerances, the lack of sufficient lubrication and poor bearing surface materials, MEMS with bearing surfaces experience considerable wear and fail after prolonged testing [57]. Efforts to improve the bearing materials and lubricants have been partially successful [58].

4. Transduction mechanisms

A transduction mechanism is a physical effect that converts signals from one form of energy (e.g., electrical, magnetic, mechanical, thermal, chemical or radiative) into another form. Many different physical effects have been used in MEMS to transduce signals in sensors and actuators. A table of prominent transduction mechanisms used in MEMS is given in figure 15.

In sensors often more than one transduction mechanism is used in series to end up with an electrical signal (e.g.,

To From	Electrical	Magnetic	Mechanical	Thermal	Chemical	Radiative
Electrical		Ampereis Law	Electrostatics, Electrophoresis	Resistive Heating	Electrolysis, Ionization	EM Transmission
Magnetic	Hall Effect, Mag. Resistance		Magnetostatics, Magnetostriction	Eddy Currents Hysteretic Loss	Magnetic Separation	Magneto-optics
Mechanical	Variable Cap. Piezoresistance Piezoelectricity	Magnetostriction		Friction	Phase Change	Tribo- luminescence
Thermal	Thermoelectric	Curie Point	Thermal Expansion		Reaction Rate Ignition	Thermal Radiation
Chemical	Electrochemical Potential	Chemomagnetic	Phase Change	Combustion		Chemo- luminescence
Radiative	Photoconductor, EM Receiving	Magneto-optics	Radiation Hardening	Photothermal	Photochemical	

Figure 15. Table of common transduction mechanisms used in MEMS.

in micromechanical magnetometers the magnetic signal is converted to a mechanical signal and then from mechanical to electrical). In actuators, transduction mechanisms are typically used in series to convert an electrical signal into a mechanical signal (e.g., in thermal actuators an electrical signal is converted to a thermal signal and then from thermal to mechanical). Specific examples of MEMS that use some of these transduction mechanisms are given in the following sections.

5. Microsensors

Micromachining and MEMS are technologies well suited to improve the performance, size and cost of sensing systems. For this reason the greatest commercial successes in MEMS are microsensors and they represent the majority of MEMS developed to date. Although historically, the greatest demand and most research and development activity has been on pressure sensors and accelerometers, the field of MEMS is maturing and the diversity of applications and sensor technologies has increased tremendously. Although it is impossible to describe each microsensor technology in any detail, the most prominent microsensor technologies are described below.

5.0.1. Strain gauges. Strain gauges are used to transduce a mechanical signal into an electrical signal. Traditionally, this is accomplished by measuring the change in resistance of a strained metallic conductor. However, the change in resistance of a semiconductor material, such as silicon, can produce a much larger effect (e.g., 10–100 times larger) [59]. For this reason, miniaturized silicon piezoresistors are integrated into sensors that require a measurement of mechanical strain (e.g. the deformation of a membrane in a pressure sensor or the deflection of a flexure attached to a proof mass in an inertial sensor).

5.0.2. Capacitive position detection. An increasingly common method of transducing a mechanical position signal into an electrical signal uses the variation in capacitance caused by electrode motion [60]. Advantages of capacitive position detecting over piezoresistive measurements include smaller size, higher sensitivity and lower power. Although in some cases conventional parallel-plate capacitance configurations are used to detect vertical motion, the freedom and precision of micromachining can be used to form a common design that

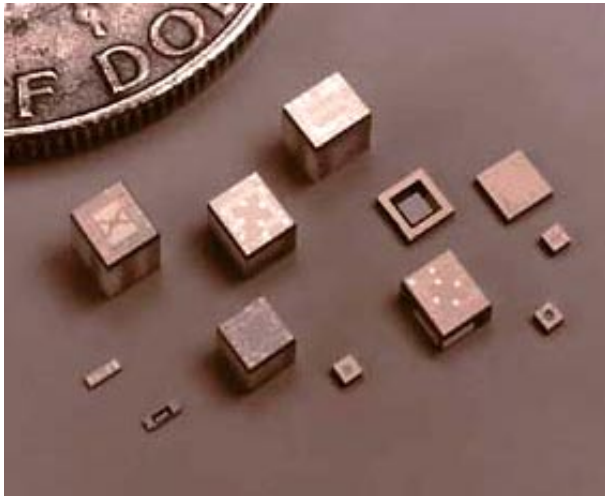


Figure 16. Micromachined pressure sensor dice with the smallest having dimensions $175 \times 700 \times 1000 \mu\text{m}^3$ [61].

uses arrays of interdigitated electrodes that sense changes in lateral position (i.e., in-plane motion).

5.1. Pressure sensors

A good example of a commercially successful low-cost disposable medical pressure sensor developed by Lucas NovaSensor NPC-107 is shown in figure 16 [61]. In it a silicon micromachined sensing element is used to meet or exceed industry requirements (e.g., sensitivity within $\pm 1\%$ and linearity better than 1%). The smallest of the micromachined silicon pressure sensors shown in figure 16 is made small enough ($1 \text{ mm} \times 0.7 \text{ mm} \times 0.175 \text{ mm}$ thick) to be placed in the tip of a catheter and inserted into blood vessels.

These microsensors are produced by combining bulk-micromachining techniques with wafer bonding to form small but robust and reliable micromembranes. Although the piezoresistors are formed in a Wheatstone bridge to maximize signal transduction, active electronics are typically not integrated with these sensors to save cost. Pressure sensors have also been made with a surface micromachining process and use capacitive position detection to measure the deflection of a small membrane (e.g., $290\text{--}550 \mu\text{m}$ in diameter) [62].

5.2. Inertial sensors

Many different inertial microsensors have been made (e.g., single- and multi-axis accelerometers and gyroscopes) using either piezoresistors or capacitive position detection. Good examples of the capacitive design are the accelerometers produced by Analog Devices Inc. (figure 17).

Note that the sensor and circuits are monolithically integrated onto one chip. This is done to minimize any parasitic stray capacitance that would otherwise significantly degrade the signal (e.g., bond wires are a large source of unstable stray capacitance) [63].

5.3. Magnetometers

Historically most microfabricated magnetometers have been completely solid state (i.e., no moving parts). Examples in-

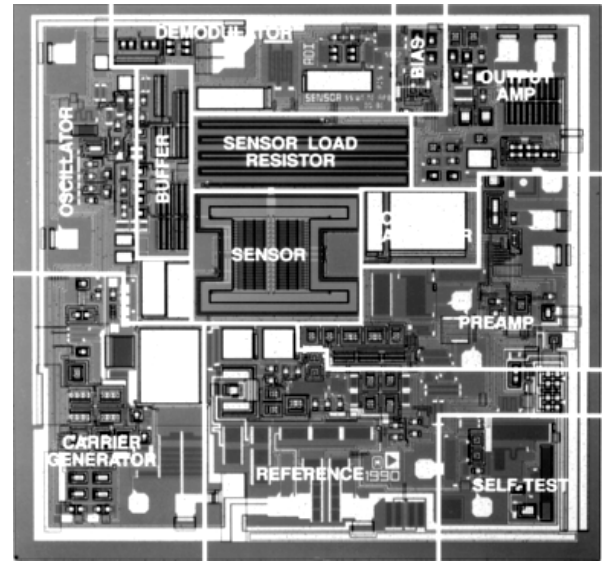


Figure 17. Monolithic accelerometer with capacitive position detection (Analog Devices, Inc) [63, 64].

clude semiconductor-based magnetometers [65, 66] (e.g., Hall effect, magnetodiodes and magnetotransistors), magnetoresistive sensors [65, 67], flux-gate magnetometers [68, 69] and SQUIDs [65]. Recently new magnetometer designs have been proposed and developed that involve Lorentzian-force-generated mechanical resonance or magnetostatically induced motion that are proportional to the sensed field. Although the motion is precisely detected by optical methods, eventually capacitive detection can be used to realize an integrated chip-scale solution [70].

5.4. Thermal sensors

The thermal isolation of MEMS can be tailored by controlling the design of the thermal conductance of their mechanical supports (e.g., material composition, length-to-width aspect ratio and length-to-thickness aspect ratio). The integration of a temperature-sensing element onto the thermally isolated region can be used to quantify the microscopic temperature fluctuations. An array of such elements has been used to quantify the amount of incident infrared thermal energy and has enabled the production of inexpensive room-temperature thermal imagers [71, 72].

5.5. Chemical sensors

Chemical sensors represent a rapidly growing segment of the microsensor market. The primary applications are as *in vitro* diagnostic instruments, drug screening, genetic screening, implantable sensors and environmental monitoring. Although the micromachining of chemical sensors is typically simple, other components sometimes used in a complete chemical sensor system (i.e., sample preparation and delivery, reaction control and waste disposal) are more difficult to integrate together.

Examples of relatively simple chemical microsensors include those which have an electrical impedance that varies as a function of gas composition and concentration. The resistive

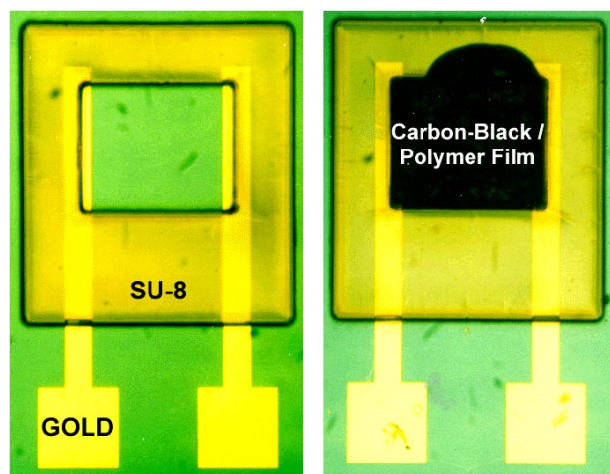


Figure 18. Microfabricated polymer carbon-black gas sensor with SU-8 microwell for solvent containment [74].

films that transduce chemical signals into electrical signals include conductive polymers, polymers doped with conductive particles and some metal oxides. The challenges common to impedance-based chemical sensors include identifying single gases, quantifying gas concentration, dealing with gas mixtures, sensitivity to water vapor, sensitivity to temperature changes, microfabrication of arrays of uniquely sensitive sensors and integration with circuits.

5.5.1. Polymer-based gas sensors. Many polymers will geometrically swell reversibly when exposed to certain gases. To use insulating polymers, they are doped with conductive particles to reduce their impedance (e.g., carbon black). When doped, the overall resistance of the doped polymer will change as a function of the chemically specific and concentration-dependent swelling [73]. One difficulty is that the polymers will swell to a greater or lesser extent when exposed to a variety of gases. To identify specific gases, the response pattern of many different polymers is needed.

In order to microfabricate arrays of sensors with unique polymers, the integration process must contend with the large volume of solvent that is typically present during polymer deposition. Furthermore, the microfabrication technique must not damage previously deposited polymers. Another strategy is to use a permanent microwell structure to contain the polymer-solvent solution in a well defined sub-millimeter area without disturbing previously deposited polymers. An example of a polymeric impedance-based gas sensor that uses an SU-8 microwell structure is given in figure 18.

Metal oxides. The conductivity of certain metal oxides, most commonly SnO_2 , is known to vary as a function of the concentration of specific gases (e.g., O_2 , H_2 , CO , CO_2 , NO_2 and H_2S) when the metal oxide is heated sufficiently to induce a chemical reaction that is detected. There are several mechanisms that cause the resistance of the metal oxide to vary [4, 75].

Resonant sensors. The resonant frequency of a mechanical element is strongly dependent on its geometry, mechanical

properties and mass. By coating a resonating mechanical element, such as a beam or membrane, with a compound that will selectively bind to only specific ions or molecules, the mass of the mechanical element will increase when they are present. The ion-concentration-dependent mass loading can be determined by measuring the corresponding shift in the resonant frequency.

Common resonant chemical sensors use either acoustic waves driven along surfaces of a solid plate or in a thin membrane (i.e., surface acoustic waves, SAWs, and flexural plate waves, FPWs, respectively) or the shift in resonance of a coated cantilever beam [76, 77]. Acoustic-wave sensors have been used to detect liquid density, liquid viscosity and specific gas vapors. Design challenges for resonant sensors include (1) temperature sensitivity of the mechanical flexure, (2) selectivity of the binding compound and (3) reversibility of the binding and mass loading process.

5.5.2. Electrochemical sensors. The oxidation and reduction of chemical species on a conducting electrode can be observed by measuring the movement of charge. There are two primary methods of sensing electrochemical reactions: potentiometric and amperometric. Potentiometric sensors can be used to measure the equilibrium potential established between the electrode material and the solution, a potential that is dependent on the chemistry involved. Amperometric sensors measure the current generated by a reaction and thus give a measure of reaction rates. By controlling the potential of the electrode relative to the solution and measuring the resultant charge flow induced, the presence of specific ions can be determined by observing the potential at which they undergo oxidation or reduction. This is a process known as voltammetry.

Micromachining processes can be used to accurately and reliably define the area, number and relative position of electrodes that are exposed to solution. In addition, the simple construction of a typical electrochemical sensor (i.e., a partially insulated metal trace on a substrate) allows ICs to be easily integrated with the electrode. The ICs can be used to provide on-chip signal processing and amplification.

ISFETs. Field effect transistors (FETs) are very sensitive to variations in the amount of charge on their controlling electrodes (i.e., gates). If an ionic solution acts as the gate of a FET, the device will be tremendously sensitive to the overall ion concentration of the solution (i.e., not selective to specific ions). A good pH sensor can be made this way and indeed one exists [78]. By coating the gate of the FET with a compound that will selectively bind or allow to pass only specific ions or molecules, an ion-sensitive FET, or ISFET, can be realized. Common difficulties with ISFETs, as with all chemical sensors, are drift and repeatability.

5.6. Biosensors

Sensors based on the characteristics of biological molecules and organisms can achieve extremely high specificity and can take advantage of built-in natural sensing mechanisms.

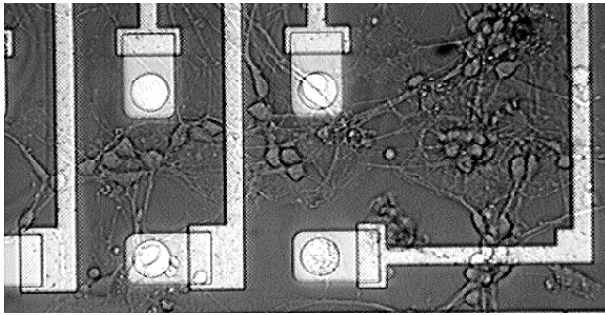


Figure 19. Cell-based biosensor with microelectrode array [81].

5.6.1. Molecular-specific sensors. Chemical sensors that respond only to specific ions or molecules can be extremely selective. Among the most selective are the interactions between complex organic molecules, such as antigens and antibodies. One caveat is that often extremely selective sensors are also less reversible and thus may require special packaging to protect the sensors until they are needed [77]. In addition, so-called GeneChips are used to detect specific strands of DNA and will be discussed in a subsequent section. A prominent example of a molecularly sensitive amperometric sensor is one that uses a glucose oxidase enzyme to detect glucose [79]. The enzyme, which is typically immobilized on or near electrodes, reacts with glucose and alters the local pH, oxygen concentration and hydrogen peroxide concentration—events that can be electrochemically detected.

5.6.2. Cell-based sensors. New innovative microsensors use living cells as the primary transduction mechanism. An advantage of using cells to detect chemicals is that cells are microscopic chemical laboratories that can amplify a chemical signal (i.e., the detection a few molecules can lead to the production of many so called ‘second messenger’ molecules)—essentially providing biological gain [80]. The amplified cell signal can be monitored by either detecting a chemical change within the cell or inferring the change by monitoring other parameters, such the electrical activity. One sensor uses chick myocardial cells to detect the presence of epinephrine, verapamil and tetrodotoxin in the cell’s environment (figure 19) [81]. Limitations of cell-based sensors include the lifetime and robustness of the cells.

5.6.3. Sensors for neural systems. The implications of MEMS technologies for neuroscience are revolutionary. We now have the potential to develop arrays of microsystems, which can be tailored to the physical and temporal dimensions of individual cells. Neuroscientists can now realistically envision sensing devices that allow real-time measurements at the cellular level. With MEMS technology, many electrodes can be co-fabricated onto a single substrate so that both precise temporal and spatial information can be obtained. MEMS technology can also be used to shape the substrate into either arrays of microprobes capable of penetrating neural tissue (figure 20) [82–86], or into a perforated membrane through which regenerating neural tissue can grow and then be monitored [87]. Information from such sensors could be monitored, analyzed and used as a basis of experimental

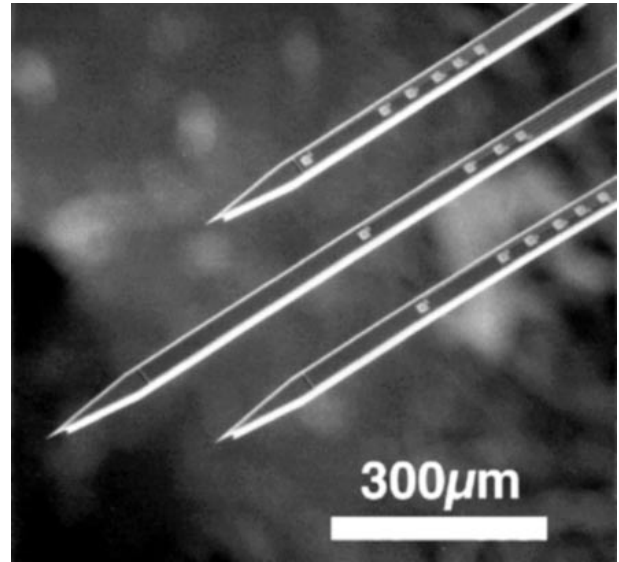


Figure 20. Microfabricated silicon neural probe arrays [82].

or medical intervention, again at a cellular level. Another example is the use of micromachined neural sensors and stimulators to control prosthetic limbs with pre-processed signals recorded from the brain or spinal column.

5.7. Summary of microsensors

Although microsensors are the most mature application of MEMS, they continue to improve and diversify. Significant on-going efforts are improving the performance of multi-axis accelerometers and gyros as well as new types of micromechanical magnetometer and biochemical sensor array. Although the standardization remains a stumbling block, groups within the IEEE are working to eliminate these obstacles. A good example of such progress is the new standard for smart sensors (i.e., IEEE 1451).

6. Microactuators

The development of microactuators is less mature than that of microsensors because of the initial lack of appropriate applications and the difficulty to reliably couple microactuators to the macroscopic world. Although the earliest microactuators were driven by electrostatic forces, devices now exist that are driven by thermal, thermal phase-change, shape-memory alloy, magnetic and piezoelectric forces to name a few. Each method has its own advantages, disadvantages and appropriate set of applications. The following sections discuss each briefly and gives examples.

6.1. Electrostatic microactuators.

Electrostatic microactuators have been constructed out of metal or heavily doped semiconductors and designed with flexures, rotary bearing surfaces and linear bearing surfaces. Although electrostatic forces are proportional to the square of the applied voltage, typically tens to hundreds of volts are needed to generate enough force (e.g., a few μN) to achieve actuation on the order of a few micrometers. Improved designs

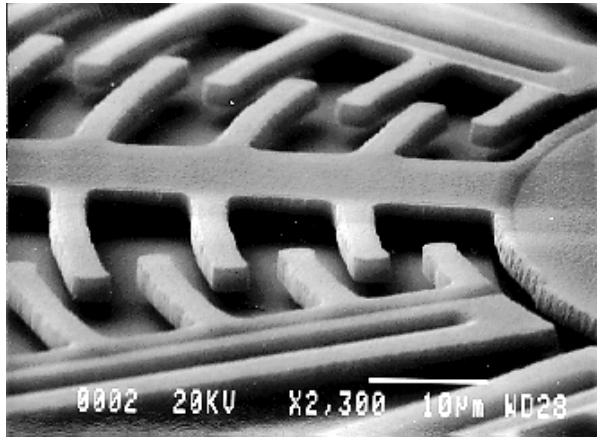


Figure 21. Electrostatic comb drive [89].

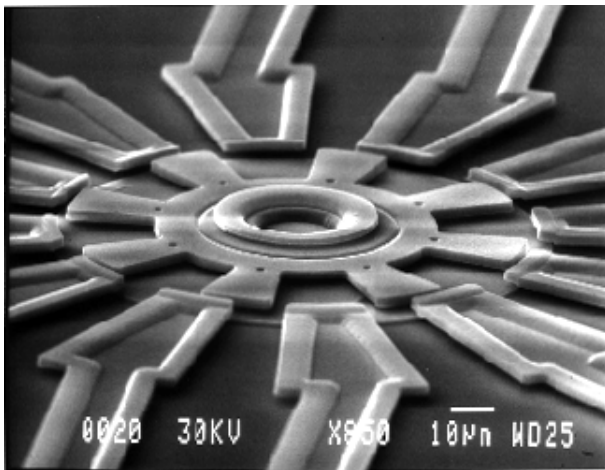


Figure 22. Rotary electrostatic micromotor [30].

can increase the range of motion by an order of magnitude [88]. The two best known designs are the lateral comb drive (figure 21) [89] and the rotary micromotor (figure 22) [30].

The large driving voltage prevents electrostatic microactuators from being conveniently driven with typical on-chip circuits and voltages (e.g., <5 V). In addition, high voltages on small features create tremendous electric field gradients that attract dust particles. Also, electrostatic actuators will not function in conductive fluids. A more recent electrostatic actuator design that can achieve very large forces ($\sim$ mN) and can travel long distances (6 mm) is an electrostatic *scratch-drive* stepper motor [90]. This highly area efficient design has been used to assemble complex hinged three-dimensional microstructures that are used for optical applications [56].

6.2. Thermal microactuators.

Early thermal microactuators are straightforward bi-morph designs that take advantage of a considerable difference in the thermal expansion coefficient of each material in the device [4]. A clever design can achieve a similar extent of actuation with the nonsymmetric heating of a single layer of patterned material [91]. Another unique design takes advantage of the considerable force created during the phase change from liquid to vapor [92]. Also, a special class of materials known as

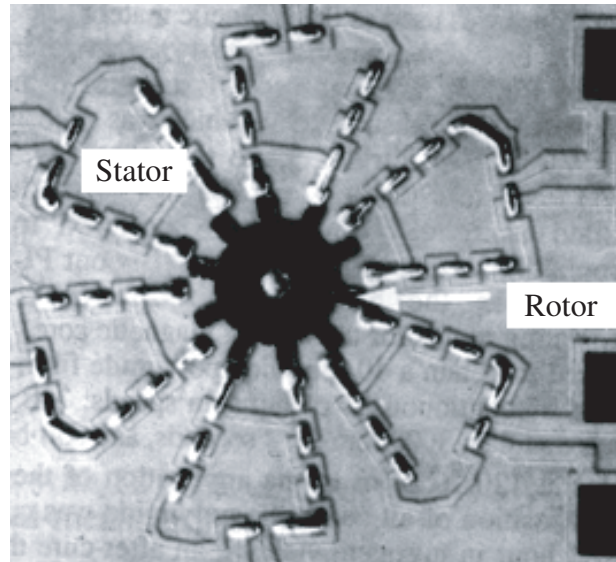


Figure 23. A magnetic microactuator designed as a miniaturized version of an electromagnetic motor [95].

shape-memory alloys can undergo a radical change in shape and size when heated. This thermally driven phase change from a low-temperature and weak martensite crystal phase to a higher-temperature and very rigid austenite crystal phase can be exploited for microactuation [93, 94]. The advantage of shape-memory alloys is that they can generate a large force and stroke for a relatively small change in temperature.

6.3. Magnetic microactuators.

Early magnetic microactuators followed conventional macroscopic designs, despite the significant challenge involved in integrating ferromagnetic cores, rotors and copper coils around the cores (figure 23) [95, 96]. As discussed earlier, magnetic microactuators can achieve larger forces over larger gaps than their electrostatic counterparts. Magnetic actuation is a more robust actuation mechanism than electrostatics, because it can operate in conductive fluids and the lower electric field gradients present in magnetic microactuators will not attract appreciable quantities of dust particles. The price is higher design and processing complexity.

Other magnetic actuator designs do away with the costly need to integrate coils by using off-chip sources for the magnetic field. Although such magnetic microactuators can only generate torques, they can easily generate large out-of-plane deflections (e.g., more than 90°)—useful for microphotonic and millimeter wave applications (figure 24) [97–99]. In addition, the short-range but high-force displacements generated by magnetostrictive materials (i.e., materials that experience a mechanical strain when magnetized) have been used in microactuators with a bi-morph construction.

6.4. Piezoelectric microactuators.

Actuators that use piezoelectric materials generate large forces over small displacements and thus are also typically used in a bi-morph or multi-layer construction. The high-frequency

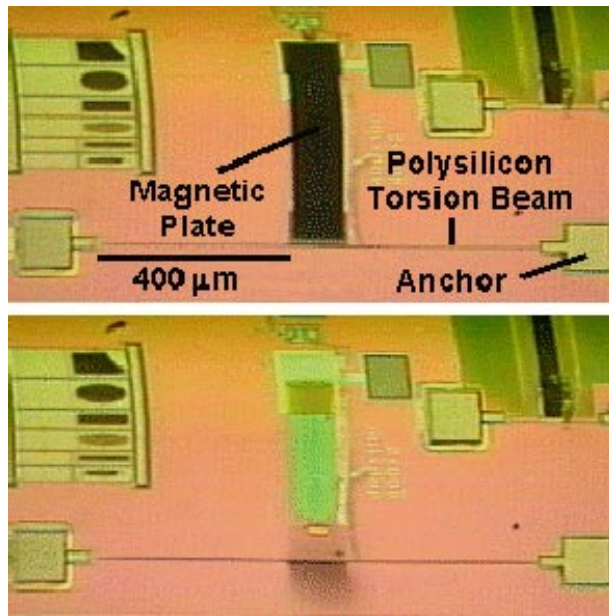


Figure 24. Magnetic microactuators that use off-chip field [97–99].

response of piezoelectric materials (e.g. ZnO and PZT) enables the small repeated displacements to rapidly accumulate when configured in a stepper motor design. These characteristics of piezoelectric microactuators have been used, for example, in surgical applications (e.g., a smart force-feedback knife [100], and an ultrasonic cutting tool [101]).

6.5. Summary of microactuators

As the research and development of microactuators matures, the devices are finding broader application and acceptance by the marketplace, even though their development is still time consuming and costly. A good example is the extremely broad effort to develop arrays of microactuated optical components for all-optical networks and wavelength-division-multiplexing (WDM) systems. It is estimated that currently there are more than 30 companies in the US and more than 50 companies internationally working to develop microphotonic systems. Although not all are using a microactuator-based technology, nearly all are taking advantage of micromachining. Since most microactuators are custom developed for specific applications, no microactuator standards yet exist.

7. Microsystems

Microsystems consist of microsensors (to quantify the inputs), circuits (to determine an action based on sensor input) and microactuators (to effect the computed change). Ideally the power source and communication components would also be miniaturized to the same scale. Communication with the microsystem to determine its state of operation and to change operating parameters can be done with a direct connection or wirelessly [102, 103]. The microfabrication of an entire system represents the ultimate accomplishment of the on-going revolution in miniaturization. Presently, there are very few microsystems that meet this definition.

One example of a chip-scale microsystem is the ADXL50 accelerometer by Analog Devices, Inc. This closed-loop microsystem uses capacitive displacement detection to measure the motion of the proof mass, integrated circuits to determine the voltage necessary to balance the inertial motion and electrostatic actuators to control the position of the proof mass.

However, the most prominent microsystems to date are those that control fluids on a microscale to enable micro total analysis systems (μ TAS). Typically, these systems consist of microreservoirs, microchannels, microvalves, micropumps, microfilters and detectors. For practical and economic reasons, circuits are typically not monolithically integrated with the microfluidic systems. Examples are described in the following sections.

7.1. Micro total analysis systems (μ TAS)

The ability to electrically control fluid flow in micromachined channels (i.e. pumping and valving) without any moving parts has enabled the realization of micromachined complex chemical analysis [104]. With multiple independently controlled fluid flow, complex sample preparation, mixing and testing procedures can be realized. Electrically controlled electro-osmotic flow or electrophoretic flow are the two most common methods used to control microscale fluids [105]. Liquid chromatography (i.e., a method of separating liquids based on their different mobility in a long flow channel) can be used to perform a precise chemical analysis in microfabricated flow channels. Sensors integrated at the end of the flow channel can reveal a time-domain spectrum of the fluid composition. Micromachined electrophoretic devices have been used to separate ions and DNA molecules ranging in size from 70 to 1000 bases in under 2 min—much faster than conventional macroscale capillary electrophoresis systems [4, 106]. The detection of each ion or molecular species can be accomplished with electrochemical measurements, fluorescence or optical absorption.

7.2. Microsystems for genetic analysis

The analysis of genetic material typically involves first the amplification of a DNA sample and then its detection. The amplification of a DNA sample can be accomplished by a polymerase chain reaction (PCR), which is a process that begins by heating the DNA sample above the temperature at which the two strands separate or ‘melt’ ($\sim 90\text{--}95^\circ\text{C}$). If the DNA polymerase enzyme and the building blocks of DNA (i.e., nucleotide triphosphates) are present during cooling, the DNA polymerase will then reconstruct each double helix, resulting in a doubling of the number of DNA stands. A major advantage of miniaturizing PCR systems is the fact that the much lower thermal mass of the reaction chamber allows for more rapid heating and cooling and thus a much faster cyclic process overall (figure 25) [107]. Furthermore, integrating heaters and temperature sensors into the same chip could allow for more precise temperature control.

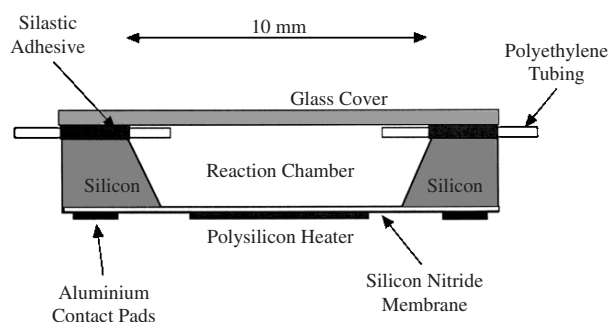


Figure 25. Micromachined PCR chamber [107].

7.2.1. Gene chips. Although separation by electrophoresis can be used to detect the size distribution of DNA molecules, another method is needed to determine their precise code. One method used to determine the code exploits the highly selective hybridization process that binds DNA fragments to only their complementary sequence. In order to test for many specific sets of DNA sequences (i.e., for genetic screening), a large number of unique oligonucleotide probes need to be constructed and compared with the amplified DNA.

One novel method of constructing oligonucleotide probes employs the same lithographic techniques as used to construct MEMS. Specifically, a substrate is coated with a compound that is protected by a photochemically cleavable or photolabile protecting group (e.g., nitroveratryloxy-carbonyl). When this film is exposed to a pattern of light, the illuminated regions become unprotected and can be conditioned to receive a specific nucleotide paired with a photolabile protecting group. By continuing the processes with a new mask pattern each time, very large arrays of unique combinations of nucleotide can be rapidly formed. The process is repeated until the desired oligonucleotides are constructed. After tagging the sample DNA with a fluorescent probe, it is then distributed over the array of microscale oligonucleotide probes. Subsequent optical inspection of the distribution of fluorescence clearly indicates which oligonucleotides in the array match with a portion of the sample DNA. Miniaturization of this detection system allows massively parallel screening (i.e., 40 000 different compounds can be tested on a single 1 cm² chip with 50 μm oligonucleotide probe areas). Affymetrix, Inc, has commercialized a DNA detection scheme based on this technology [108].

7.3. Micropumps, microvalves, microfilters and microneedles

In order to produce completely miniaturized microfluidic systems, it may also be necessary to miniaturize the components needed to pre-process and post-process the fluids (e.g., micropumps [109], microvalves [61] and microfilters [110]), [4].

7.3.1. Microvalves. Several different types of microvalve have been microfabricated, including normally open and normally closed valves, and can be used to control gases or fluids. Furthermore, several actuation mechanisms have been used to generate the large forces needed in microvalves: thermal (HP, NovaSensor), thermal phase change

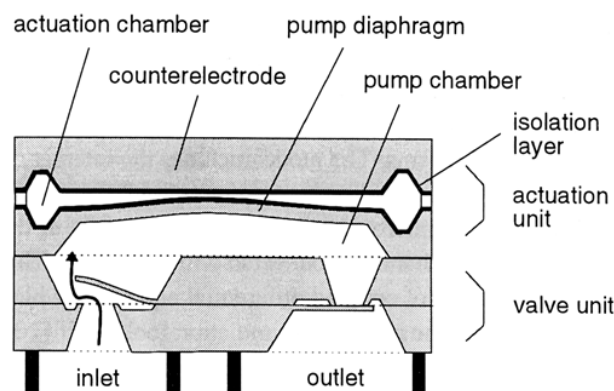


Figure 26. Electrostatic micropump with two one-way check valves (from [109]).

(Redwood Microsystems [111]), shape-memory alloy (TiNi) and magnetic (University of Cincinnati) [4].

The performance of microvalves compares favorably with macroscopic solenoid valves. In particular, microvalves typically operate faster and have a longer operational lifetime. However, since microvalves are typically driven by thermal transduction mechanisms, their power consumption is relatively high (0.1–2.0 W). Care must be taken to prevent the valve temperature from exceeding that tolerated by the fluid or gas media being controlled.

7.3.2. Micropumps. Similarly, several methods of microactuation have been used to drive micropumps: electrostatic forces [109], magnetic forces (MEMSTek [112]) and piezoelectric. One example is an electrostatically driven micropump produced by bonding multiple bulk micromachined silicon wafers together. The bonding process creates a pumping cavity with a deformable membrane and two one-way check valves. The electrodes are fabricated inside a second isolated cavity formed above the deformable pumping membrane so that they are sealed away from the conductive solutions being pumped (figure 26). Although the micropump works well, high voltages (> 100 V) are required for significant pumping to occur.

When designing micropumps for biomedical applications, attention must be paid to the media being pumped. Some fluids, such as insulin, cannot tolerate aggressive mechanical pumping mechanisms without degrading.

7.4. Summary of microsystems

The miniaturization of a complete microsystem represents one of the greatest challenges to the field of MEMS. Reducing the cost and size of high-performance sensors and actuators can improve the cost performance of macroscopic systems, but the miniaturization of entire high-performance systems can result in radically new possibilities and benefits to society. Microfluidic systems constitute the majority of present microsystem development efforts due to their broad applicability, particularly as bio-chemical analysis systems (e.g., diagnostic tools). As microactuator technology matures, the number and diversity of microsystems will also increase.

8. MEMS CAD

The existence of commonly available computer-aided design (CAD) software for ICs and has enabled accurate simulations to be performed instead of costly and time-consuming microfabrication and electrical testing. Completing several design iterations in the time it would take to microfabricate one design has saved the IC industry considerable amounts of money and speeds up the overall product design process.

Similarly capable CAD software for MEMS is expected to have the same large impact. However, the problems being solved in MEMS are far more diverse and thus require more solvers and a more complex (and likely expensive) CAD tool. To compound the difficulties, the solvers must actually solve for a coupled solution since each mechanism typically interacts. For example, when an electrostatically actuated microflexure deflects in response to an applied voltage, the charge on the flexure and the ground plane will immediately change to reflect this mechanical motion. The result is that the net force on the flexure will increase. Existing solvers compute the equilibrium solution in a lengthy iterative process. Ideally, the MEMS CAD tool would be capable of rapidly solving mechanical, thermal, electrostatic, magnetic, fluidic, RF and optical solutions in a coupled fashion.

Another challenge is the complete simulation of MEMS that are integrated with circuits. Often the MEMS component is simulated separately to develop a very accurate behavioral model. However, a new model must be generated each time the geometry is changed. Recently a concerted effort is being made to develop techniques for decomposing an integrated MEMS design into a hierarchical design of basic MEMS elements or standard MEMS cells [113]. The benefit is that such a design can be fully integrated into the IC CAD tool. In this case, the MEMS and IC solutions are solved for simultaneously by the same tool and changes to the size of the basic elements can be immediately determined.

9. Commercialization of MEMS

Enormous commercial opportunities exist for MEMS products [114]. In fact, in 2000 a few MEMS companies commercializing arrays of optical cross connect switches for fiber-optic communication systems were purchased for \$750M (Cronos) to \$1.3B (Xros). Since MEMS are often essentially miniaturized versions of existing commercially successful products, a careful analysis of the costs involved with developing and manufacturing a MEMS-based alternative must be performed. It is typically not enough that the device is smaller and performs better; it must also be considerably cheaper. Otherwise the systems integrator will not take the risk to switch to an unproven MEMS product. Yet, due to the potentially tremendous advantages of MEMS, most existing companies producing non-MEMS solutions must do a careful analysis of the scalability of their products. The major problem with MEMS commercialization is the lack of sufficient standards for the packaging and interface (e.g., electronic, mechanical, fluidic, magnetic, optical, chemical etc). Without the standards, it is difficult for the field to offer cost-minimized solutions guaranteed to work in existing systems.

10. New opportunities (MEMS in space)

There is a strong push to use MEMS on future space missions, both in orbital satellites and inter-planetary exploration modules, due to all of their potential advantages (e.g. smaller size, lower weight, lower power consumption) [115]. In addition to reducing the size of on-board instrumentation, MEMS technology has been used to produce micropropulsion systems. The Jet Propulsion Laboratory (JPL) is one organization in particular that is championing the use of MEMS in space, particularly in future missions to Mars. The Aerospace Corp. is also pushing the development of so-called pico-satellite technology (i.e., satellites smaller than $\sim 1 \text{ dm}^3$). An important concern is the impact of high levels of radiation for extended periods of time on the performance of MEMS. Experiments with an integrated inertial MEMS device have demonstrated that the effect of radiation can be considerable [116].

11. To find out more

The best source of information on the development of MEMS technology used to be the primary MEMS-dedicated conferences

- *MEMS. IEEE Micro Electro Mechanical Systems Workshop* held in late January every year and rotates between the US, Europe and Asia.
- *Transducers. International Conference on Solid-State Sensors and Actuators* held in June on odd-numbered years and rotates between the US, Europe and Asia.
- *Hilton Head. Solid-State Sensor and Actuator Workshop* held in June on even-numbered years on Hilton Head Island.
- and journals
- *IEEE Journal on Microelectromechanical Systems*
- *Sensors and Actuators* (Physical A and Chemical B)
- *Micromechanics and Microengineering*;

however, with the spread of MEMS technology into other fields (e.g., optics, bioengineering etc), the conferences of these other fields have created symposia dedicated to the application of MEMS technology to their specific field. Although this has rapidly increased the acceptance and spread of MEMS technology, its wide breadth makes it difficult to remain informed on all the latest developments.

12. Conclusions

The potential exists for MEMS to establish a second technological revolution of miniaturization that may create an industry (or industries) that exceeds the IC industry in both size and impact on society. Micromachining and MEMS technologies are powerful tools for enabling the miniaturization of sensors, actuators and systems. In particular, batch fabrication techniques promise to reduce the cost of MEMS, particularly those produced in high volumes. As the field of MEMS is adopted by many disciplines and various advantageous scaling properties are exploited, the diversity and acceptance of MEMS will grow. Reductions in cost and increases in performance of

microsensors, microactuators and microsystems will enable an unprecedented level of quantification and control of our physical world. Although the development of commercially successful microsensors is generally far ahead of the development of microactuators and microsystems, there is an increasing demand for sophisticated and robust microactuators and microsystems. Supporting the growing development of new MEMS technologies are on-going efforts to improve MEMS standardization as well as hierarchical MEMS CAD solutions that can be integrated with IC CAD tools for full real-time system-level simulations. Despite the challenges involved in commercializing MEMS, the number and scale of such commercialization efforts are growing rapidly. In fact, in the near future MEMS may play a tremendously significant role in the never-ending exploration of space.

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